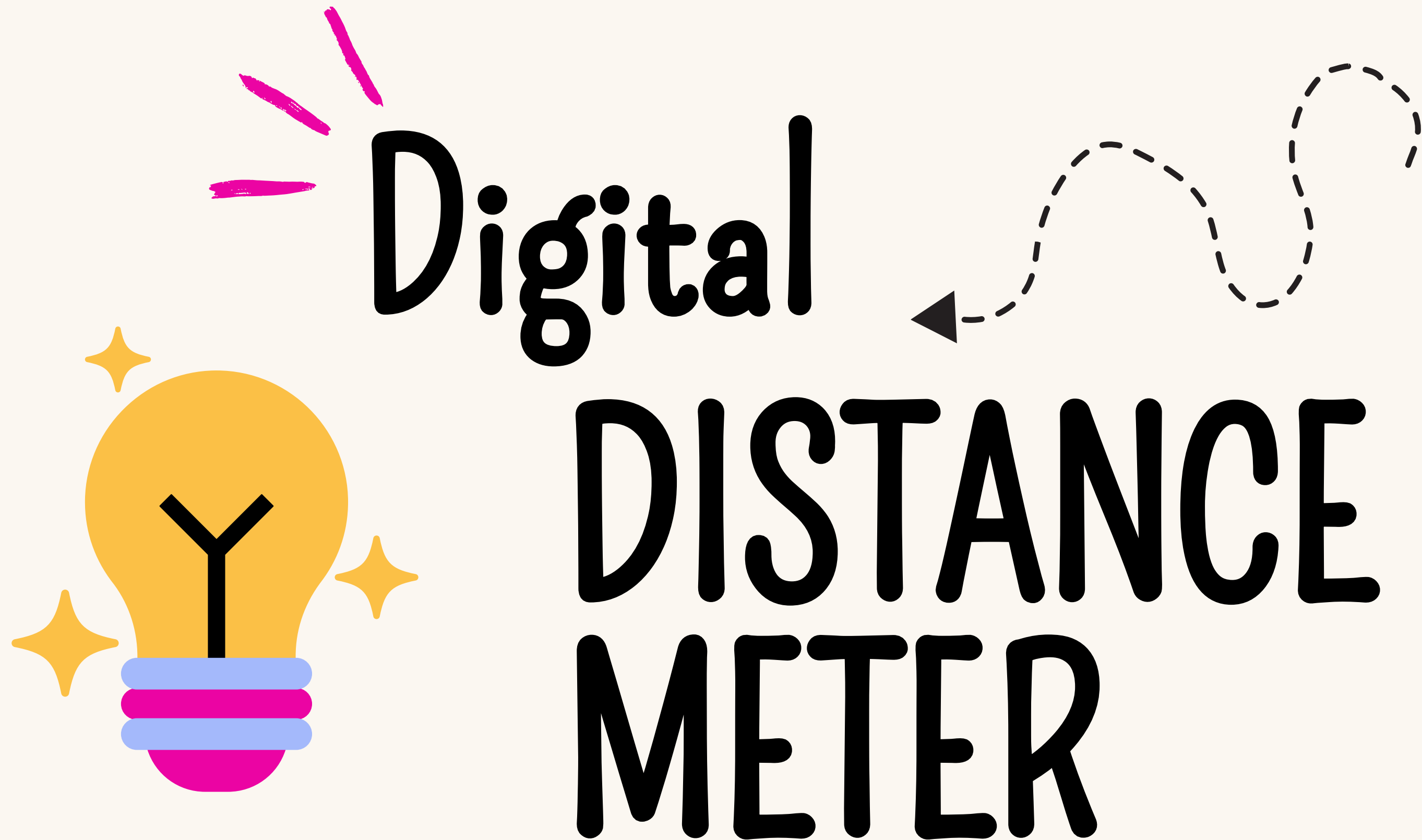
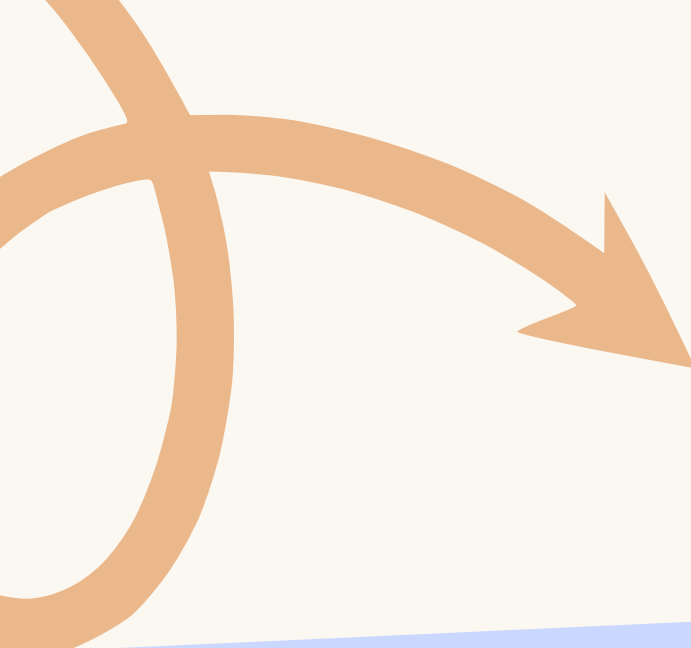


Members : Batoul Atris, Nagham  
Krayem , and Razane Hijazi



# Digital DISTANCE METER

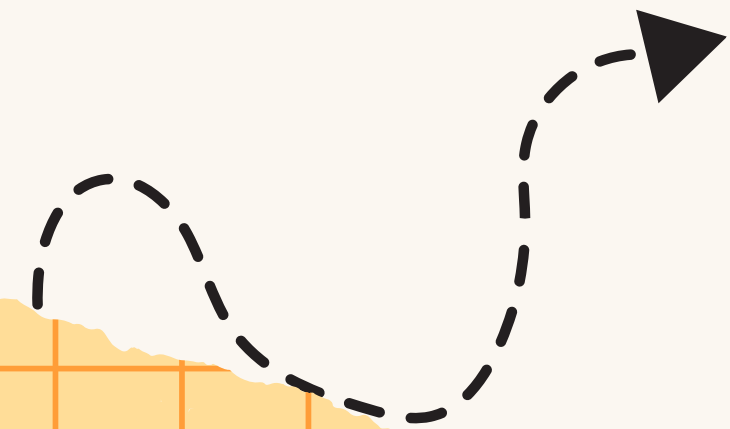
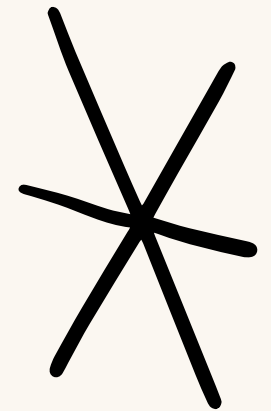
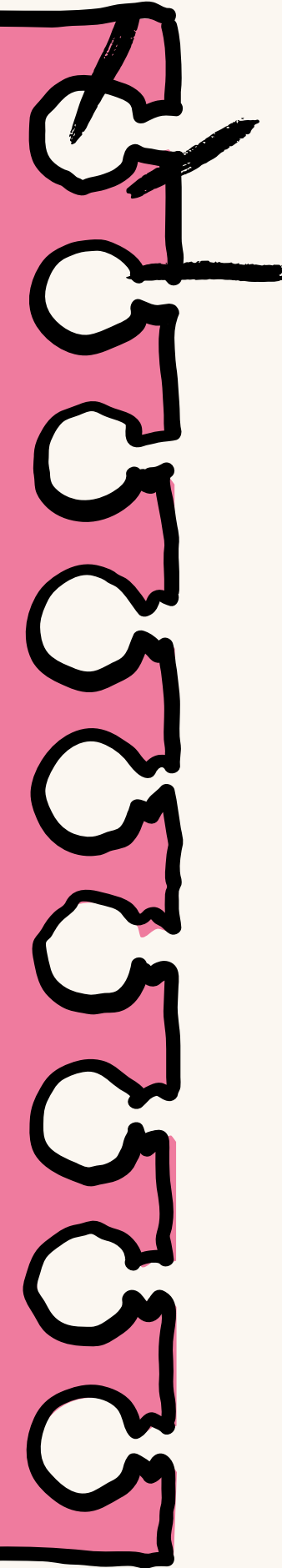


So:

1. What are the objectives from making it ?
2. What are the components used ?
3. How does it work ?



1. Our purpose is to measure the distance separating the element from the circuit ,by showing the measurements on LCD screen with the ability to detect the area by an angle of range 180 degree , in addition to that having the ability to alert if there is a too close subject to it.



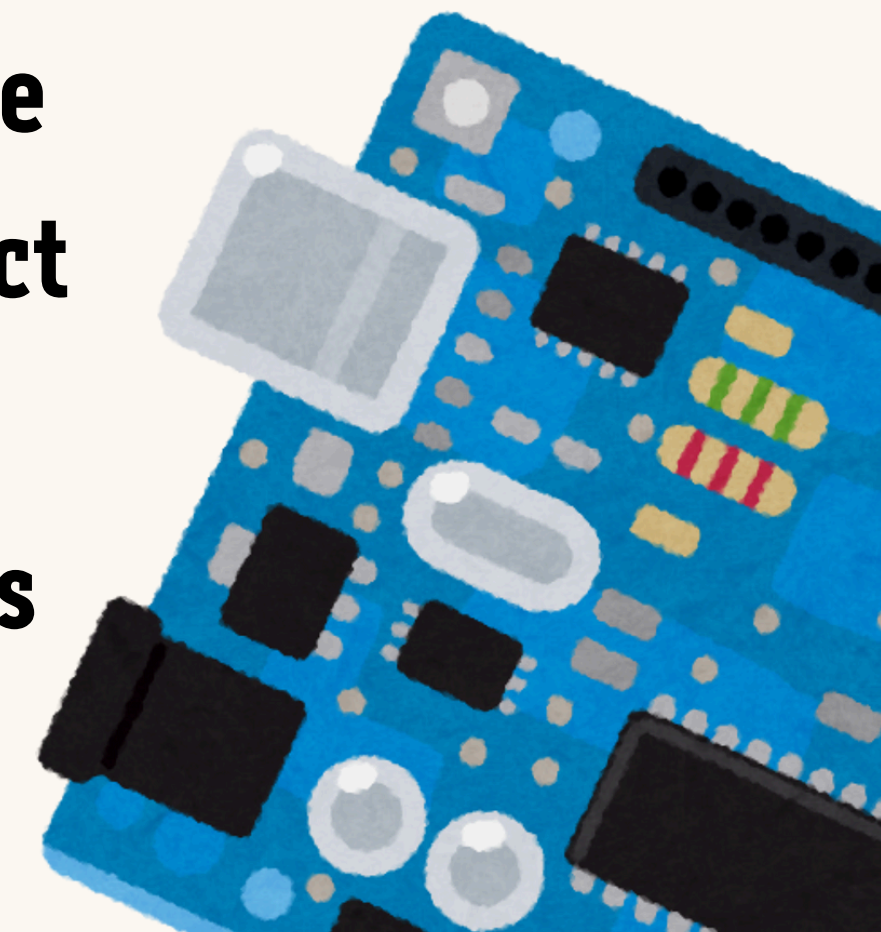


## **Components**

- 1 Arduino UNO R3**
- 2 Servo Motor**
- 3 LCD screen**
- 4 Buzzer**
- 5 Ultrasonic Sensor**
- 6 Jumper wires male/male , female/male**
- 7 Bread Board**
- 8 Solar panel 5V 0,3W**
- 9 Battery 3.7 V**
- 10 Switch**

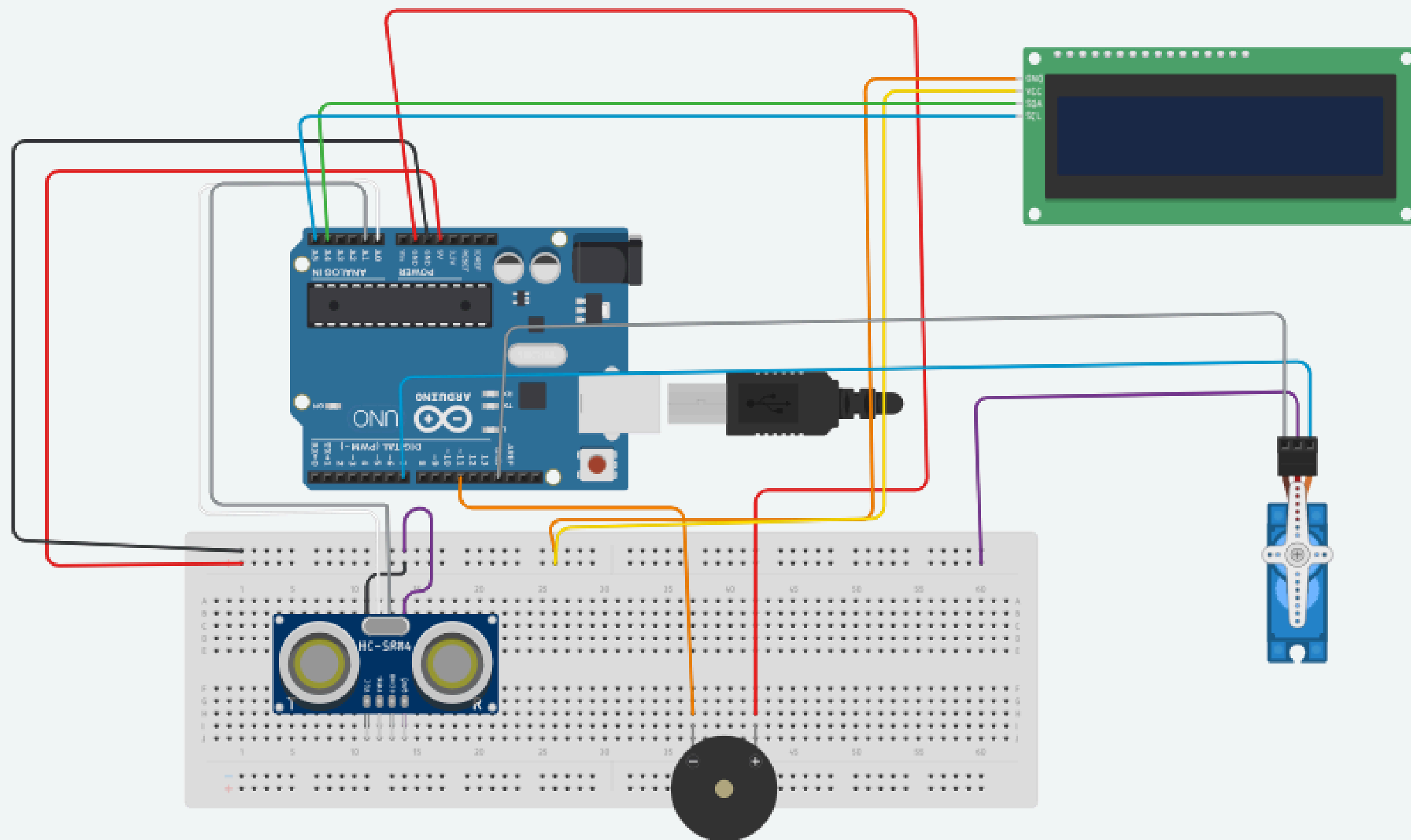
**These components were chosen by the team members after a whole resaerch about the project , these components were convenient in the size , needed power ( value of voltage ) , the performance , and finally the cost**

**Where Arduino is the “ BRAIN OF THE CIRCUIT ” it is a programmable board [ programmed by us according to our purpose by Arduino IDE app] , and all the mentioned components are connected to it through the jumper wires and the bread board (it is just such extra pins ), then when the sensor detects a subject it starts measuring the separating distance between them , with a periodic rotating {half rotation 180 deg} servo motor , where at the same time all the measurements are being displayed on the LCD screen [ the distance and the angle of the servo motor] .As well as , if this moving subject come close to the sensor “closer than 10 cm” the buzzer starts alerting by forming sound similar to “tootle” until the subject goes away from this specified close area .**



# CIRCUIT DIAGRAM 1

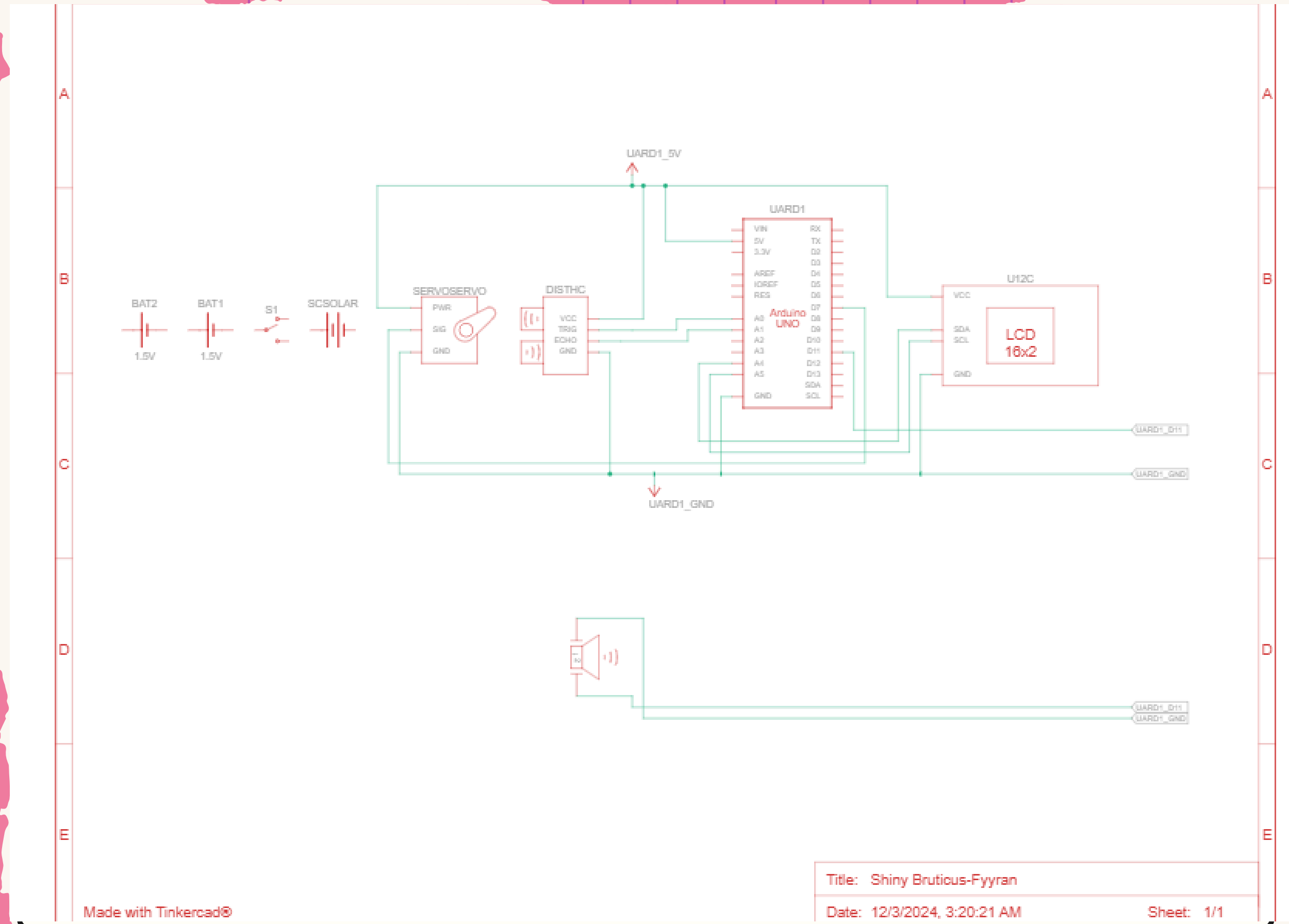
BY  
TINKERCAD





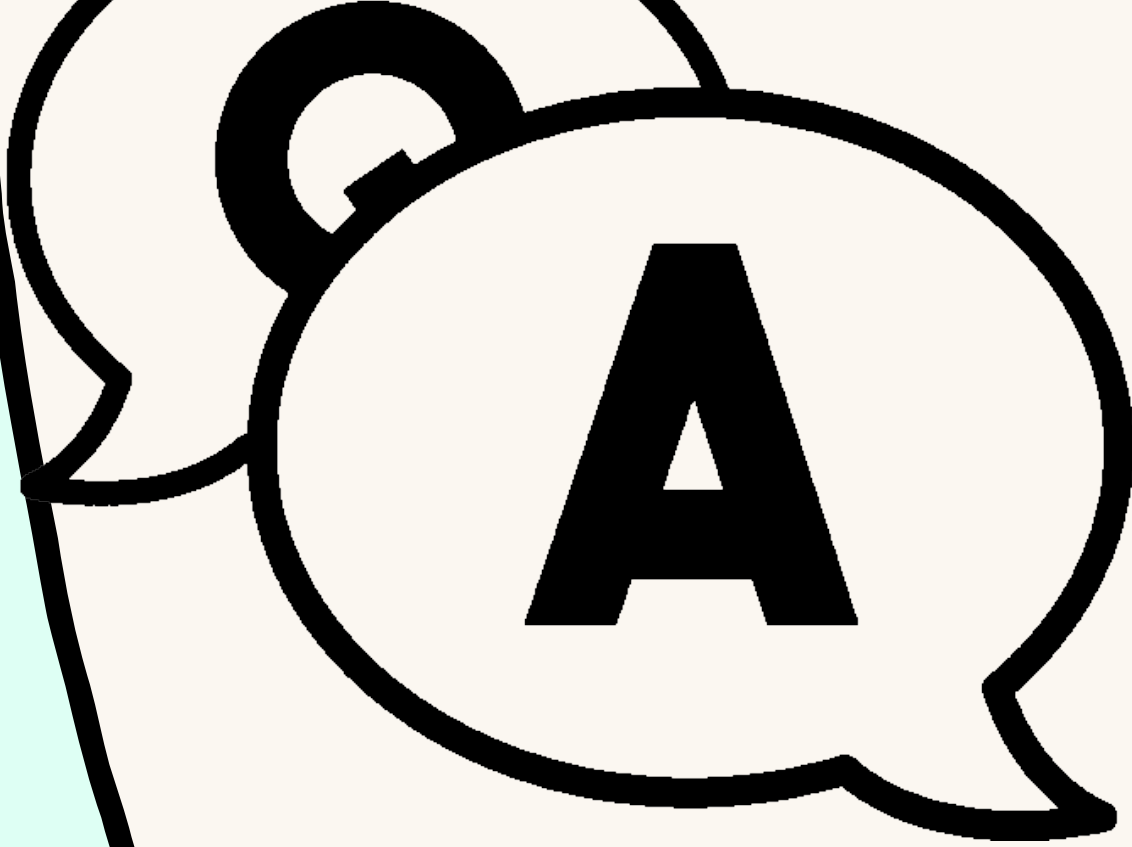
# CIRCUIT DIAGRAM 2

BY  
TINKERCAD





**Remark : We did not put the connections of the solar panel , battery , switch , because there are too missing components on TINKERCAD , such as the charging module etc .. We are trying to explain them in the real -life Project .**



**THANKS FOR  
FOLLOWING AND  
WATCHING !!!**